

Michael C. Anoruo

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EDUCATION

Carnegie Mellon University

Doctor of Philosophy in Robotics

Advisor: Wennie Tabib, Resilient Intelligent Systems Lab (RISLab)

Pittsburgh, PA

Anticipated May 2029

University of Maryland, Baltimore County

Bachelor of Science in Computer Science — AI/ML Track

GPA: 3.93/4.0, Summa Cum Laude

Baltimore, MD

Dec 2023

GRADUATE COURSES

Learning for 3D Vision	Deep Reinforcement Learning
Optimal Control & Reinforcement Learning	Robot Learning
Mobile Robots	Computer Vision
Math Fundamentals for Robotics	Natural Language Processing
Data Science	

UNDERGRADUATE COURSES

Machine Learning	Intro to AI
Algorithm Design & Analysis	Data Structures
Operating Systems	Software Engineering I
Linear Algebra	Calculus I–II

SKILLS

Languages: Python, C++, C, Java, Lua

Robotics & Hardware: ROS, ROS 2, Vicon, Crazyflie, KiCad

Deep Learning & RL: PyTorch, TensorFlow, Gymnasium

Computer Vision: OpenCV, Open3D, PyTorch3D

Scientific Computing: NumPy, SciPy, Matplotlib, Pandas

DevTools: Docker, Git, Linux

Build Tools & Automation: CMake, Colcon, Maven

Web & Databases: Flask, Apache, PostgreSQL

PHD RESEARCH

Carnegie Mellon University — Resilient Intelligent Systems Lab (RISLab)

Ph.D. Student

Pittsburgh, PA

Aug 2024 – Present

- Developed a differentiable, GPU-accelerated quadrotor simulator in PyTorch capable of simulating 100k+ drones in parallel.
- Built a Gymnasium environment to train RL quadrotor control policies for dynamic target interception, achieving over 95% success against targets moving up to 10 m/s.
- Equipped Crazyflie drones with onboard vision by soldering third-party monocular cameras and integrating a receiver to stream live image feeds to desktop.
- Modernized the lab's Crazyflie flight stack by setting up and configuring the drones to interface with the lab's Vicon motion-capture system, enabling swarming capability that wasn't previously possible.
- Deployed an MLflow experiment-tracking server (PostgreSQL-backed, hosted as an Apache service) to centralize training runs and provide on-demand access to results for a 5-person research team.

RESEARCH EXPERIENCE

Johns Hopkins University APL — Asymmetric Operations Sector

Software Engineer Intern

Baltimore, MD

May 2024 – Aug 2024

- Enhanced a GUI module and contributed to an integration layer in Java/JavaFX for a secure communication system, interfacing with encryption APIs to transmit data as part of a large engineering team.
- Led a team of 7 interns on a drone swarming project for search-and-rescue, researching and testing A*-based search algorithms in PyBullet simulation.
- Trained a variational autoencoder on network packet data to support anomaly-based network intrusion detection.

- Built an internal logging service (PostgreSQL, Flask REST API, web interface) for network monitoring, saving the company hundreds of thousands of dollars in Splunk licensing fees.

University of Maryland, Baltimore County — DPrime.ai Lab
Data Scientist Intern

Baltimore, MD
Dec 2022 – Dec 2023

- Built preprocessing scripts to segment task-evoked pupillary response (TEPR) time-series data by detecting significant deviations from mean pupil diameter.
- Generated labeled pupil-response segments across four cognitive tasks (attention, mental arithmetic, vigilance, working memory), enabling a teammate to train cognitive-load classification models.
- Developed a real-time visualization tool to display live pupil diameter and other biometric sensor data as it was collected during experiments.

Massachusetts Institute of Technology — Space Enabled Media Lab
Zero Robotics Research Intern

Cambridge, MA
Jun 2022 – Aug 2022

- Optimized PID controller parameters for the MIT Astrobe robot simulator.
- Developed Python scripts to automate data collection and analysis for the Zero Robotics project.
- Collaborated with a team of six, alongside NASA astronauts, to host the Zero Robotics 2022 technical competition.

University of Maryland, Baltimore County — Ebiquity Lab
Cybersecurity Research Intern

Baltimore, MD
Sep 2020 – May 2022

- Designed a DQN-based reinforcement learning algorithm to evade malware classifiers by modifying file headers, identifying detection vulnerabilities in classifier models.
- Annotated CVE records with recommended mitigation actions using spaCy, contributing labeled data to train an NLP model for automated security-phrease detection and classification.
- Contributed to an adversarial reinforcement learning environment in CyberBattleSim, where attacker and defender agents learned to counter each other, guided by a cybersecurity knowledge graph — resulting in a paper published at IEEE ICMLA 2022.

University of Pennsylvania — Scalable Autonomous Robots Lab
Robotics Research Intern

Philadelphia, PA
Jun 2021 – Aug 2021

- Collaborated with a PhD student to bring the lab's Clearpath Heron USV back online, testing the bathymetric sensor and validating autonomy behavior by collecting and replaying ROS bagfiles from field experiments.
- Repaired the lab's miniature autonomous boats, wiring motors, integrating OptiTrack motion capture for localization, and flashing onboard Arduino controllers to test collaborative object-pushing behaviors.
- Designed a waypoint algorithm to generate a strategic path for the lab's Autonomous Surface Vehicle (ASV) to survey the Schuylkill River.
- Developed a Python simulation framework to model ASV experiments.
- Created a PCB with KiCad to translate RS232 to USB for data transmission.

PROFESSIONAL EXPERIENCE

Apple Inc. — Apple Service Engineering
Data Engineer Intern

Seattle, WA
May 2023 – Aug 2023

- Replaced manual video segmentation with an automated CV pipeline, enabling accurate per-segment viewership reporting to sponsors.
- Developed code utilizing computer vision and machine learning techniques to automate analysis of large volumes of video content, cutting analysis time from over 3 hours to under 10 minutes.

ACADEMIC PROJECTS

Interactive 3D Scene Reconstruction

UMBC | Fall 2023

- Adapted PyTorch's ResNet and MiDaS pretrained models to detect objects and their 3D position from a 2D image.
- Developed a robust web scraping algorithm to query and download open-sourced 3D models from SketchFab.
- Utilized the Panda3D engine to reconstruct a 2D picture into a 3D low-fidelity virtual simulation.

Song Genre Prediction

UMBC | Fall 2023

- Curated a custom dataset of song lyrics and genre using the Spotify and Genius Lyrics APIs.

- Preprocessed lyrics, removing common characteristic words to prevent bias in model predictions.
- Developed a Recurrent Neural Network model in PyTorch to predict song genre based on its lyrics.

PERSONAL PROJECTS

Roblox Game Development

Sep 2021 – Aug 2023

- Served as lead developer for a fast-paced FPS game built in Roblox Studio, coding core game mechanics, weapons, effects, and animations.
- Designed and developed in-game menus, HUDs, shop/inventory systems, leaderboards, and settings interfaces.
- Led a small team of 3 developers, growing the game to nearly 1 million players in collaboration with a YouTuber with 10 million subscribers.

PUBLICATIONS

- Dang, Q., Kucukosmanoglu, M., **Anoruo, M.**, Kargosha, G., Conklin, S., & Brooks, J. "Automatic Detection of Cognitive Events Using Machine Learning and Understanding Models' Interpretations of Human Cognition." *Scientific Reports* 15, no. 1 (2025): 30506.
- Piplai, A., **Anoruo, M.**, Fasaye, K., Joshi, A., Finin, T., & Ridley, A. "Knowledge Guided Two-Player Reinforcement Learning for Cyber Attacks and Defenses." *2022 21st IEEE International Conference on Machine Learning and Applications (ICMLA)* (2022): 1342–1349.

CONFERENCES / PRESENTATIONS

NVIDIA Day Robotics Research Talk , Carnegie Mellon University	Pittsburgh, PA, May 2026
Apple Service Engineering Intern Spotlight , Apple Inc.	Seattle, WA, Aug 2023
MSRP Summer Research Symposium , Massachusetts Institute of Technology	Cambridge, MA, Aug 2022
On-Ramp Mission Research Symposium , National Security Agency	Baltimore, MD, Mar 2022
Undergraduate Research and Creative Achievement Day , UMBC	Baltimore, MD, Apr 2022
SUNFEST Symposium , University of Pennsylvania	Philadelphia, PA, Aug 2021

TEACHING ASSISTANT EXPERIENCE

Carnegie Mellon University	Pittsburgh, PA
Teaching Assistant — 16-761 Mobile Robots	Spring 2026
Teaching Assistant — 16-362 Mobile Robot Algorithms Laboratory	Fall 2025
• Held office hours, graded assignments, and answered student questions on Piazza. • Set up the Crazyflie drone platform for use in the Mobile Robot Algorithms Laboratory.	

LEADERSHIP

Project Lead , AI Scholars Program, CMU	Summer 2026
Resident Assistant , Residential Life, UMBC	Fall 2022 – Fall 2023
Site Lead , Creative Coders, UMBC	Spring 2023
Peer Advisor , Meyerhoff Scholars Program, UMBC	Fall 2022 – Spring 2023
Executive Board , National Society of Black Engineers, UMBC	Fall 2021 – Spring 2023

ACTIVITIES

Member , System Administration & Software Development Club, UMBC	Fall 2020 – Spring 2023
Member , Institute of Electrical and Electronics Engineers, UMBC	Fall 2020 – Spring 2023
Member , Cyber Dawgs Competitive Cybersecurity Team, UMBC	Fall 2020 – Spring 2022

AWARDS and HONORS

GEM Fellow	Fall 2024
Google CSRMP Scholar	Spring 2022
NIDA EDUCATE Scholar	Fall 2021
Meyerhoff Scholar	Fall 2020
National Football Foundation Scholar Athlete	Spring 2020

VOLUNTEER EXPERIENCE

Coordinator , Robotics Institute Merch Distribution, CMU	Spring 2026 – Present
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Youth Leader, Youth Ministry, Church Community
Team Member, Media Department, Church Community
Competition Coach, MESSA Competition, NSBE Junior
Counselor, Pathfinders Club, Church Community
Math Tutor, Arbutus Achievers Tutoring Program, UMBC

Fall 2021 – Fall 2023
Summer 2018 – Spring 2023
Fall 2021 – Spring 2022
Summer 2019 – Spring 2022
Fall 2020 – Spring 2021